



Linear Algebra Lecture 5

Matrix operations

PRIME

 TU Delft



Programme

- Matrix addition and scalar multiplication
- Matrix multiplication
- Transpose of a matrix



Polling questions using Vevox



- Go to vevox.app and enter the session ID:
- No personal informations is required



Matrix addition and scalar multiplication

Matrix addition



Let $A = \begin{bmatrix} 1 & 1 \\ 3 & 5 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & -2 \\ 4 & -1 \\ -2 & 3 \end{bmatrix}$,

compute $A + B$ if possible.

A. $\begin{bmatrix} 3 & -1 \\ 7 & 4 \\ -2 & 3 \end{bmatrix}$

B. $\begin{bmatrix} 2 & -2 \\ 5 & 0 \\ 1 & 8 \end{bmatrix}$

C. Not defined



Matrix addition



Let $A = \begin{bmatrix} 1 & 1 \\ 3 & 5 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & -2 \\ 4 & -1 \\ -2 & 3 \end{bmatrix}$,

compute $A + B$ if possible.

A. $\begin{bmatrix} 3 & -1 \\ 7 & 4 \\ -2 & 3 \end{bmatrix}$

B. $\begin{bmatrix} 2 & -2 \\ 5 & 0 \\ 1 & 8 \end{bmatrix}$

✓ C. Not defined



Matrix addition and scalar multiplication



Let $A = \begin{bmatrix} 1 & 1 \\ 3 & 5 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & -2 \\ 4 & -1 \end{bmatrix}$,

compute $2A - B$ if possible.

A. $\begin{bmatrix} 0 & 4 \\ 2 & 11 \end{bmatrix}$

B. $\begin{bmatrix} -2 & -6 \\ -2 & 12 \end{bmatrix}$

C. Not defined



Matrix addition and scalar multiplication



Let $A = \begin{bmatrix} 1 & 1 \\ 3 & 5 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & -2 \\ 4 & -1 \end{bmatrix}$,

compute $2A - B$ if possible.

✓ **A.** $\begin{bmatrix} 0 & 4 \\ 2 & 11 \end{bmatrix}$

B. $\begin{bmatrix} -2 & -6 \\ -2 & 12 \end{bmatrix}$

C. Not defined



Some special types of matrices

Definitions:

An $m \times n$ matrix A is called a:

- zero matrix if all entries are zeros.
- square matrix if $m = n$.

The main diagonal of a square matrix consists of the entries a_{ii} .

A square matrix is called $a(n)$:

- diagonal matrix if all off-diagonal entries are zeros.
- identity matrix if its a diagonal matrix with 1's on the main diagonal.
- lower/upper triangular matrix if all entries above/below the main diagonal are zeros.



Properties of matrix addition and scalar multiplication

Theorem:

Let A , B and C be matrices of the same size and let r and s be scalars, then:

- a. $A + B = B + A$
- b. $(A + B) + C = A + (B + C)$
- c. $A + 0 = A$
- d. $r(A + B) = rA + rB$
- e. $(r + s)A = rA + sA$
- f. $r(sA) = (rs)A$



Matrix multiplication

Composition of linear transformations

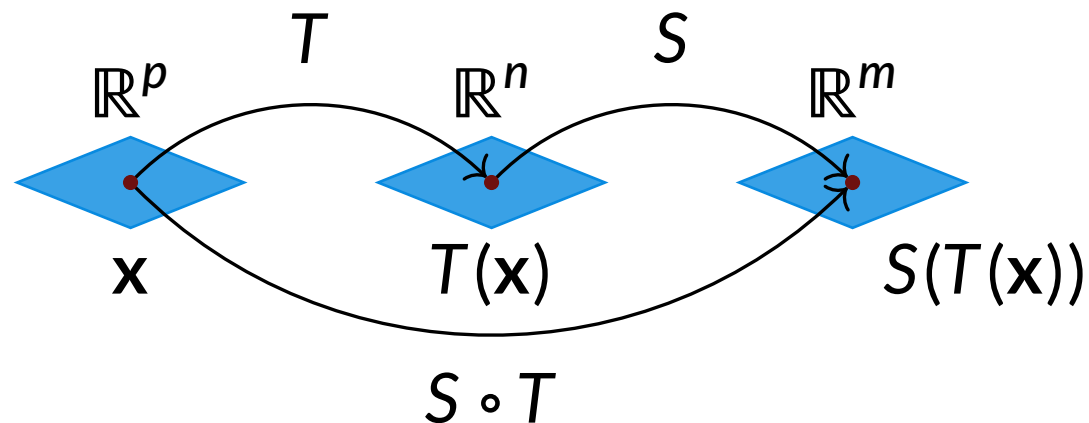
10

Theorem:

Given two linear transformations $T : \mathbb{R}^p \rightarrow \mathbb{R}^n$ and $S : \mathbb{R}^n \rightarrow \mathbb{R}^m$, then the composition $S \circ T : \mathbb{R}^p \rightarrow \mathbb{R}^m$, defined by

$$(S \circ T)(\mathbf{x}) = S(T(\mathbf{x})),$$

is also a linear transformation.



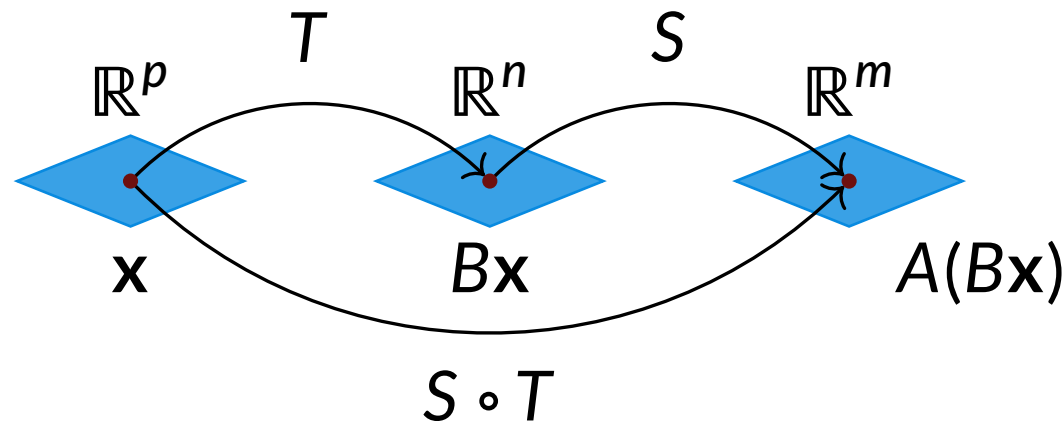
Matrix multiplication

11

Definition:

Let A be an $m \times n$ matrix and B an $n \times p$ matrix. The **product** AB of A and B (in that order) is equal to the standard matrix of the composite mapping $S \circ T$, where $S(\mathbf{x}) = A\mathbf{x}$ and $T(\mathbf{x}) = B\mathbf{x}$.

(In particular, AB is an $m \times p$ matrix).





Function composition

12

Given T and S as below. Does $(T \circ S)$ exist?

$$T\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \begin{bmatrix} x_1 + x_2 \\ x_1 - x_2 \\ -x_1 + x_2 \end{bmatrix}$$

$$S\left(\begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}\right) = \begin{bmatrix} y_1 - y_2 \\ y_2 + y_3 \end{bmatrix}$$

A. No

B. Yes





Function composition

12

Given T and S as below. Does $(T \circ S)$ exist?

$$T\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \begin{bmatrix} x_1 + x_2 \\ x_1 - x_2 \\ -x_1 + x_2 \end{bmatrix}$$

$$S\left(\begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}\right) = \begin{bmatrix} y_1 - y_2 \\ y_2 + y_3 \end{bmatrix}$$

A. No

✓ B. Yes





Matrix Multiplication

13

$S : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ is linear with matrix A .

$T : \mathbb{R}^4 \rightarrow \mathbb{R}^2$ is linear with matrix B .

Which statement on AB is correct?

- A.** AB does not exist, because $S \circ T$ does not exist.
- B.** AB does not exist, because $T \circ S$ does not exist.
- C.** AB does exist, and it is a 3×4 matrix.
- D.** AB does exist, and it is a 4×3 matrix.





Matrix Multiplication

13

$S : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ is linear with matrix A .

$T : \mathbb{R}^4 \rightarrow \mathbb{R}^2$ is linear with matrix B .

Which statement on AB is correct?

- A. AB does not exist, because $S \circ T$ does not exist.
- B. AB does not exist, because $T \circ S$ does not exist.
- ✓ C. AB does exist, and it is a 3×4 matrix.
- D. AB does exist, and it is a 4×3 matrix.



Questions?



Recall the matrix-vector product

15

Definition:

The product of a matrix A with n columns and a vector $\mathbf{x}$ with n entries is defined by

$$\begin{bmatrix} \mathbf{a}_1 & \mathbf{a}_2 & \cdots & \mathbf{a}_n \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = x_1 \mathbf{a}_1 + x_2 \mathbf{a}_2 + \cdots + x_n \mathbf{a}_n.$$



Recall the matrix-vector product

15

Definition:

The product of a matrix A with n columns and a vector $\mathbf{x}$ with n entries is defined by

$$\begin{bmatrix} \mathbf{a}_1 & \mathbf{a}_2 & \cdots & \mathbf{a}_n \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = x_1 \mathbf{a}_1 + x_2 \mathbf{a}_2 + \cdots + x_n \mathbf{a}_n.$$

This can be extended to a matrix-matrix product.



Computing AB : column rule

16

Theorem:

If A is an $m \times n$ matrix, and B is an $n \times p$ matrix with columns $\mathbf{b}_1, \dots, \mathbf{b}_p$, then the product AB is the $m \times p$ matrix whose columns are $A\mathbf{b}_1, \dots, A\mathbf{b}_p$.

That is,

$$AB = A[\mathbf{b}_1 \quad \mathbf{b}_2 \quad \dots \quad \mathbf{b}_p] = [A\mathbf{b}_1 \quad A\mathbf{b}_2 \quad \dots \quad A\mathbf{b}_p]$$





Matrix multiplication

17

Compute AB if possible where

$$A = \begin{bmatrix} -5 & 3 \\ -4 & 1 \end{bmatrix} \text{ and } B = \begin{bmatrix} 1 & -1 & 2 \\ 2 & 6 & 0 \end{bmatrix}.$$

A. $\begin{bmatrix} -1 & 4 \\ -34 & 0 \end{bmatrix}$

C. $\begin{bmatrix} 1 & -6 \\ 23 & -2 \\ -10 & 8 \end{bmatrix}$

B. $\begin{bmatrix} 1 & 23 & -10 \\ -6 & -2 & -8 \end{bmatrix}$

D. Not defined





Matrix multiplication

17

Compute AB if possible where

$$A = \begin{bmatrix} -5 & 3 \\ -4 & 1 \end{bmatrix} \text{ and } B = \begin{bmatrix} 1 & -1 & 2 \\ 2 & 6 & 0 \end{bmatrix}.$$

A. $\begin{bmatrix} -1 & 4 \\ -34 & 0 \end{bmatrix}$

C. $\begin{bmatrix} 1 & -6 \\ 23 & -2 \\ -10 & 8 \end{bmatrix}$

✓ B. $\begin{bmatrix} 1 & 23 & -10 \\ -6 & -2 & -8 \end{bmatrix}$

D. Not defined





Matrix Multiplication

18

If A is a 6×5 matrix and B is a 5×2 matrix, what is the size of AB , if defined?

A. 6×5

E. 6×2

B. 5×6

F. 2×6

C. 5×2

G. 5×5

D. 2×5

H. Not defined





Matrix Multiplication

18

If A is a 6×5 matrix and B is a 5×2 matrix, what is the size of AB , if defined?

A. 6×5

✓ E. 6×2

B. 5×6

F. 2×6

C. 5×2

G. 5×5

D. 2×5

H. Not defined



Computing AB : row-column rule

19

Entry (i, j) of AB equals $\text{row}_i(A)\mathbf{b}_j$

$$(AB)_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{in}b_{nj}$$

$$\begin{bmatrix} a_{11} & \cdots & a_{1j} & \cdots & a_{1n} \\ \vdots & & \vdots & & \vdots \\ a_{i1} & \cdots & a_{ij} & \cdots & a_{in} \\ \vdots & & \vdots & & \vdots \\ a_{m1} & \cdots & a_{mj} & \cdots & a_{mn} \end{bmatrix} \begin{bmatrix} b_{11} & \cdots & b_{1j} & \cdots & b_{1p} \\ \vdots & & \vdots & & \vdots \\ b_{i1} & \cdots & b_{ij} & \cdots & b_{ip} \\ \vdots & & \vdots & & \vdots \\ b_{n1} & \cdots & b_{nj} & \cdots & b_{np} \end{bmatrix}$$





Matrix Multiplication

20

Determine $(AB)_{3,2}$ if possible,

where $A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ -1 & -2 & -3 & -4 \end{bmatrix}$

and $B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 0 \\ 1 & -1 \end{bmatrix}$

A. -2

D. 1

B. -1

E. 2

C. 0

F. Not defined





Matrix Multiplication

20

Determine $(AB)_{3,2}$ if possible,

where $A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ -1 & -2 & -3 & -4 \end{bmatrix}$

and $B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 0 \\ 1 & -1 \end{bmatrix}$

A. -2

D. 1

B. -1

✓ E. 2

C. 0

F. Not defined





Matrix multiplication

21

Given A and B as below. Does AB exist?
If so, what is the third row of AB ?

$$A = \begin{bmatrix} -1 & 7 \\ 0 & 4 \\ 3 & -2 \end{bmatrix} \quad B = \begin{bmatrix} 0 & 1 \\ -1 & 2 \end{bmatrix}$$

- A. No
- B. Yes, and the third row equals $[7 \quad 13]$
- C. Yes, and the third row equals $[-4 \quad 8]$
- D. Yes, and the third row equals $[2 \quad -1]$





Matrix multiplication

21

Given A and B as below. Does AB exist?
If so, what is the third row of AB ?

$$A = \begin{bmatrix} -1 & 7 \\ 0 & 4 \\ 3 & -2 \end{bmatrix} \quad B = \begin{bmatrix} 0 & 1 \\ -1 & 2 \end{bmatrix}$$

- A. No
- B. Yes, and the third row equals $[7 \quad 13]$
- C. Yes, and the third row equals $[-4 \quad 8]$
- ✓ D. Yes, and the third row equals $[2 \quad -1]$



Questions?



Properties of matrix multiplication

23

Theorem:

Let A be an $m \times n$ matrix, and let B and C be matrices with sizes for which the indicated sums and products are defined. Then:

a. $A(BC) = (AB)C$

b. $A(B + C) = AB + AC$

c. $(B + C)A = BA + CA$

d. $r(AB) = (rA)B = A(rB)$ for any scalar r .

e. $I_m A = A = A I_n$





Matrix multiplication

24

Given A and B as below.

Which statement is correct?

$$A = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \quad B = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

A. $AB = BA$

B. $AB = -BA$

C. None of the above.





Matrix multiplication

24

Given A and B as below.

Which statement is correct?

$$A = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \quad B = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

- A. $AB = BA$
- ✓ B. $AB = -BA$
- C. None of the above.



Warning

25

- a. The identity $AB = BA$ is not true in general.
- b. If $AB = AC$, then in general is not true that $B = C$!
- c. If $AB = 0$, then in general it is not true that $A = 0$ or $B = 0$!



Practice

26



- Matrix addition and scalar multiplication
- Matrix multiplication



§ 2.1: 5, 7, 8, 9

You are now able to



- add two matrices, if defined, and multiply a matrix by a scalar;
- calculate the product of two matrices, if defined, using three techniques;
- relate properties of a matrix product AB to properties of the individual matrices A and B .

Powers of a matrix

27

Definition:

If A is an $n \times n$ matrix and if k is a positive integer, then

$$A^k = \underbrace{AA \cdots A}_{k \text{ factors}}$$

For $k = 0$ we define $A^0 = I$.

Remark:

$$(AB)^k = \underbrace{(AB) \cdot (AB) \cdot \cdots \cdot (AB)}_{k \text{ times}}$$



Powers of a matrix

27

Definition:

If A is an $n \times n$ matrix and if k is a positive integer, then

$$A^k = \underbrace{AA \cdots A}_{k \text{ factors}}$$

For $k = 0$ we define $A^0 = I$.

Remark:

$$(AB)^k = \underbrace{(AB) \cdot (AB) \cdot \cdots \cdot (AB)}_{k \text{ times}}$$



Rotation



28

Let A be the standard matrix of the counterclockwise rotation over an angle φ . Then A^2 is given by:

A. $\begin{bmatrix} 2 \cos \varphi & -2 \sin \varphi \\ 2 \sin \varphi & 2 \cos \varphi \end{bmatrix}$

C. $\begin{bmatrix} \cos 2\varphi & -\sin 2\varphi \\ \sin 2\varphi & \cos 2\varphi \end{bmatrix}$

B. $\begin{bmatrix} \cos \varphi^2 & -\sin \varphi^2 \\ \sin \varphi^2 & \cos \varphi^2 \end{bmatrix}$

D. $\begin{bmatrix} \cos^2 \varphi & \sin^2 \varphi \\ \sin^2 \varphi & \cos^2 \varphi \end{bmatrix}$



Rotation



28

Let A be the standard matrix of the counterclockwise rotation over an angle φ . Then A^2 is given by:

A. $\begin{bmatrix} 2 \cos \varphi & -2 \sin \varphi \\ 2 \sin \varphi & 2 \cos \varphi \end{bmatrix}$

✓ C. $\begin{bmatrix} \cos 2\varphi & -\sin 2\varphi \\ \sin 2\varphi & \cos 2\varphi \end{bmatrix}$

B. $\begin{bmatrix} \cos \varphi^2 & -\sin \varphi^2 \\ \sin \varphi^2 & \cos \varphi^2 \end{bmatrix}$

D. $\begin{bmatrix} \cos^2 \varphi & \sin^2 \varphi \\ \sin^2 \varphi & \cos^2 \varphi \end{bmatrix}$



Rotation

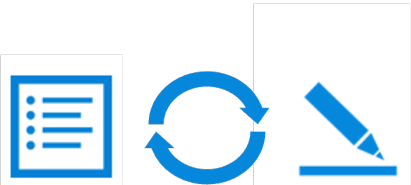
29

Is the following statement true or false?

Let A be the standard matrix of the transformation that rotates clockwise over $\frac{180}{n}$ degrees in $\mathbb{R}^2$, where $n \geq 2$ is a positive integer. Then $A^n = -I_2$.

A. True

B. False



Rotation

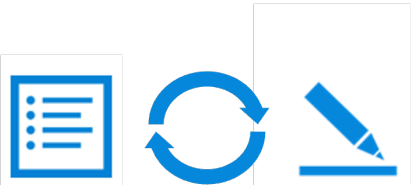
29

Is the following statement true or false?

Let A be the standard matrix of the transformation that rotates clockwise over $\frac{180}{n}$ degrees in $\mathbb{R}^2$, where $n \geq 2$ is a positive integer. Then $A^n = -I_2$.

✓ A. True

B. False



Encryption

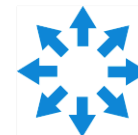
30

- Suppose you want to send an important message to one specific person so that if the message is intercepted, it can not easily be decrypted.
- Then you should make it impossible to read for others, thus encrypt the message.

A	B	C	D	E	F	G
1	2	3	4	5	6	7
H	I	J	K	L	M	N
8	9	10	11	12	13	14
O	P	Q	R	S	T	U
15	16	17	18	19	20	21
V	W	X	Y	Z		
22	23	24	25	26		

Table. Character encoding

C	O	M	P	U	T	E	R		S	C	I	E	N	C	E
3	15	13	16	21	20	5	18	0	19	3	9	5	14	3	5



Encryption

31

C	O	M	P	U	T	E	R		S	C	I	E	N	C	E
3	15	13	16	21	20	5	18	0	19	3	9	5	14	3	5

$$M = \text{Message} = \begin{bmatrix} 3 & 21 & 0 & 5 \\ 15 & 20 & 19 & 14 \\ 13 & 5 & 3 & 3 \\ 16 & 18 & 9 & 5 \end{bmatrix}$$

$$K = \text{Key} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\text{Encrypted message} = KM = \begin{bmatrix} 47 & 64 & 31 & 27 \\ 44 & 43 & 31 & 22 \\ 29 & 23 & 12 & 8 \\ 16 & 8 & 9 & 5 \end{bmatrix}$$

Encryption

32

$$\text{Key} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

A	B	C	D	E	F	G
1	2	3	4	5	6	7
H	I	J	K	L	M	N
8	9	10	11	12	13	14
O	P	Q	R	S	T	U
15	16	17	18	19	20	21
V	W	X	Y	Z		
22	23	24	25	26		

Table. Character encoding

$$\text{Encrypted message:} \begin{bmatrix} 16 & 30 & 43 & 50 & 15 & 22 & 29 & 46 \\ 15 & 14 & 30 & 30 & 15 & 21 & 20 & 30 \\ 1 & 9 & 21 & 21 & 14 & 5 & 1 & 21 \\ 1 & 9 & 1 & 7 & 0 & 0 & 1 & 7 \end{bmatrix}$$

What was the original message?

Questions?



Transpose of matrix

The transpose of a matrix

35

Definition:

For any $m \times n$ matrix A the transpose of A , denoted by A^T , is the matrix of size $n \times m$ whose columns are formed from the corresponding rows of A .

Example:

$$\begin{bmatrix} 2 & -1 \\ 3 & -2 \\ 4 & 5 \end{bmatrix}^T = \begin{bmatrix} 2 & 3 & 4 \\ -1 & -2 & 5 \end{bmatrix}$$





Matrix transpose properties

36

Given two $m \times n$ matrices A and B .

Which of the statements are correct?

(i) $(A^T)^T = A$;

(ii) $(A + B)^T = A^T + B^T$

A. (i) is true, (ii) is true,

B. (i) is true, (ii) is false,

C. (i) is false, (ii) is true,

D. (i) is false, (ii) is false





Matrix transpose properties

36

Given two $m \times n$ matrices A and B .

Which of the statements are correct?

(i) $(A^T)^T = A$;

(ii) $(A + B)^T = A^T + B^T$

✓ **A.** (i) is true, (ii) is true,

B. (i) is true, (ii) is false,

C. (i) is false, (ii) is true,

D. (i) is false, (ii) is false





Matrix product and transpose

37

For matrices

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \text{ and } B = \begin{bmatrix} 1 & 2 \\ -1 & 1 \end{bmatrix}$$

which statement is correct?

$$(i) (A^2)^T = (A^T)^2; (ii) (AB)^T = A^T B^T$$

- A.** (i) is true, (ii) is true,
- B.** (i) is true, (ii) is false,
- C.** (i) is false, (ii) is true,
- D.** (i) is false, (ii) is false





Matrix product and transpose

37

For matrices

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \text{ and } B = \begin{bmatrix} 1 & 2 \\ -1 & 1 \end{bmatrix}$$

which statement is correct?

$$(i) (A^2)^T = (A^T)^2; (ii) (AB)^T = A^T B^T$$

- A. (i) is true, (ii) is true,
- ✓ B. (i) is true, (ii) is false,
- C. (i) is false, (ii) is true,
- D. (i) is false, (ii) is false



Properties of the transpose of a matrix

38

Theorem:

Let A and B be matrices such that the operations are defined.

a. $(A^T)^T = A$

b. $(A + B)^T = A^T + B^T$

c. $(rA)^T = rA^T$ for any scalar r

d. $(AB)^T = B^T A^T$



Questions?



Practice

40



- Transpose of a matrix



§2.1: 15, 17, 19, 20, 23, 27, 29



You are now able to

- transpose a matrix

Wrap up and next lecture

41

Practice the topics of this lecture to:

- add two matrices, if defined, and multiply a matrix by a scalar;
- calculate the product of two matrices, if defined, using three techniques;
- relate properties of a matrix product AB to properties of the individual matrices A and B ;
- transpose a matrix;
- apply the calculation rules related to the aforementioned learning goals.

Next lecture:



Invertibility of matrices



Sections 2.2 and 2.3



See you next lecture

PRIME

 **TU**Delft

